

When Risk Is Not Rewarded

A Short Guide to One of the Most Puzzling Anomalies in Equity Markets



CONCRETUM RESEARCH

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Modern portfolio theory is built on a remarkably intuitive idea: **investors should earn higher expected returns for bearing greater risk.**

This principle lies at the heart of the **Capital Asset Pricing Model (CAPM)**, one of the most influential models in financial economics. According to the theory, stocks with higher systematic risk should compensate investors with higher long-term returns.

Yet among the many anomalies documented in empirical finance, **few challenge this principle more directly than the low-volatility anomaly.**

Over the past several decades, researchers have repeatedly found that stocks with the highest historical volatility have **not only delivered lower risk-adjusted returns, but often lower absolute returns as well.** In other words, investors who accepted the greatest amount of risk were frequently the least rewarded.

In our previous articles, we introduced the concept of **decile portfolios** and showed how they can be used to study well-known equity characteristics such as **Value**. In this article, we apply the same methodology to another of the most robust anomalies in equity markets: **Low Volatility**.

The results are surprisingly difficult to reconcile with traditional asset-pricing theory.



The Low-Volatility Puzzle

Back in 1972, **Fischer Black** observed that the empirical **Security Market Line (SML)** was **too flat**. High-beta stocks earned lower average returns than predicted, while low-beta stocks performed better than the CAPM suggested.

Decades later, **Andrew Ang and his co-authors documented a closely related result: stocks with high idiosyncratic volatility tended to earn abnormally low future returns (link)**. Subsequent research showed that this effect was not limited to the United States, but appeared across several developed equity markets.

The chart below applies the same decile methodology used in our previous articles. At the end of each month, stocks are ranked by their historical variance and divided into ten portfolios. Decile 1 contains the lowest-variance stocks, while Decile 10 contains the highest-variance stocks.

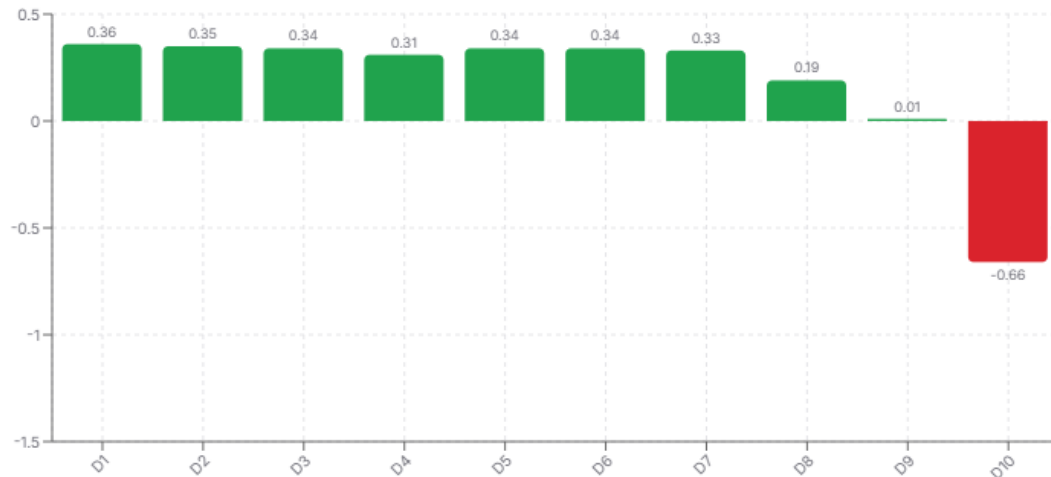
Performance Metrics — Variance

Avg Returns (%)

CAPM Alphas (%)

Sharpe Ratios

Max Drawdowns (%)



Monthly CAPM Alphas by Variance Deciles. Chart generated by [Concretum Factor Tracker](#).

The results are quite **surprising**.

There is **no positive relationship between risk and return**. In fact, the stocks with the highest variance perform worse than the rest of the market.

- The **lowest-variance portfolios generate positive CAPM alphas**.
- As variance increases, performance gradually deteriorates.
- The **highest-variance decile ends up producing a strongly negative alpha**.

This finding clearly **contradicts the basic idea behind traditional asset pricing theory**, which suggests that **investors should earn higher expected returns for taking more risk**. In our sample, we observe exactly the opposite.

The same conclusion emerges when looking at Sharpe ratios. **The highest-volatility decile produces by far the worst risk-adjusted**

performance over the whole sample period. To put some numbers on it, the lowest-variance decile achieved a Sharpe ratio of 0.71, while the highest-variance decile reached only 0.03.

These results help explain why the low-volatility anomaly is often considered one of the clearest contradictions of modern financial theory.

Why Does the Anomaly Exist?

There is no single universally accepted explanation. The most common interpretations combine institutional constraints, behavioral biases, and limits to arbitrage. Let's explore some of them...

Leverage constraints. Many institutional investors **cannot use leverage**, so they overweight high-beta stocks instead of leveraging low-beta stocks.

Behavioral biases. Investors are drawn to **high-volatility, "lottery-like" stocks**, often overpaying for them.

Short-selling constraints. The worst-performing high-volatility stocks are often difficult to short, **allowing mispricing to persist.**

How to Trade the Low Volatility Anomaly

The **low-volatility effect can be transformed into a long-short systematic strategy** by buying stocks with low volatility while shorting stocks with high volatility.

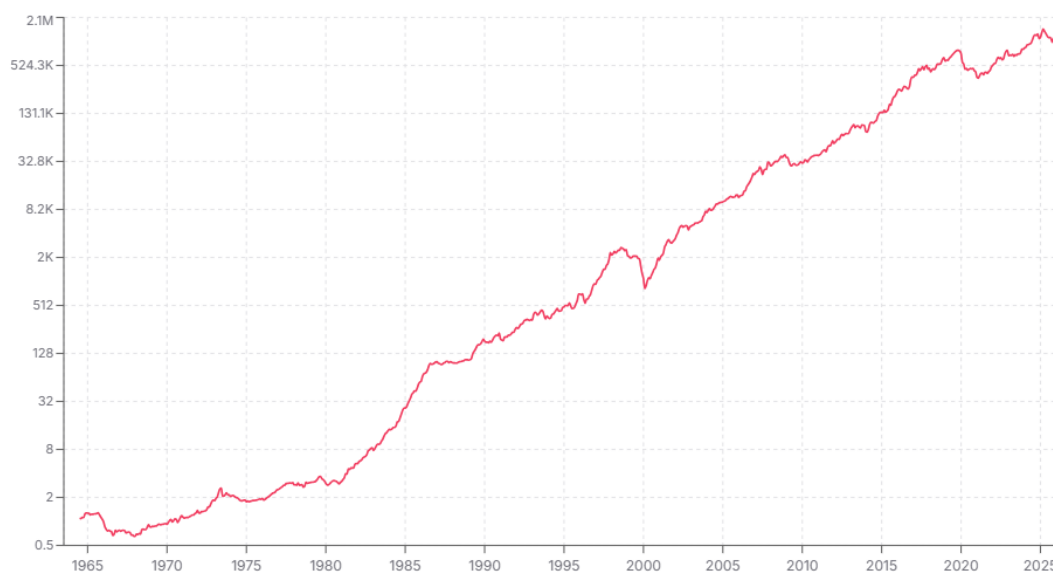
A "volatility-neutral" implementation could be constructed as follows:

- **Go long the lowest-volatility decile**
- **Short the highest-volatility decile**
- **Scale the two legs to the same target volatility (20% p.a.).**

Over the full sample **(1963–2026)**, this simple strategy would have delivered a **26% compound annual growth rate (CAGR)**, with a **Sharpe ratio of 0.87**, an **annualized CAPM alpha of approximately 21%**, and **virtually zero market beta**.

However, the strategy is **far from risk-free**. Low-volatility portfolios have historically experienced prolonged periods of underperformance, particularly during sharp market rebounds when speculative, high-beta stocks can move aggressively.

The equity curve below highlights several notable drawdown episodes, including **1966**, the **late-1990s technology bubble**, and the **post-COVID rebound**. Some of these drawdowns exceeded **50%**, illustrating that even one of the most persistent anomalies in equity markets can experience long and painful periods of adverse performance.



Equity curve of a long-short portfolio that buys the lowest-variance decile and shorts the highest-variance decile.

The Volatility Anomaly in 2026

The charts shown throughout this article are based on **equally weighted decile portfolios**, which provide the cleanest representation of the academic low-volatility anomaly.

In practice, however, many investors implement factor portfolios using **value-weighted** constituents.

Interestingly, **2026 has so far provided a useful reminder that factor premiums can experience meaningful short-term deviations.**

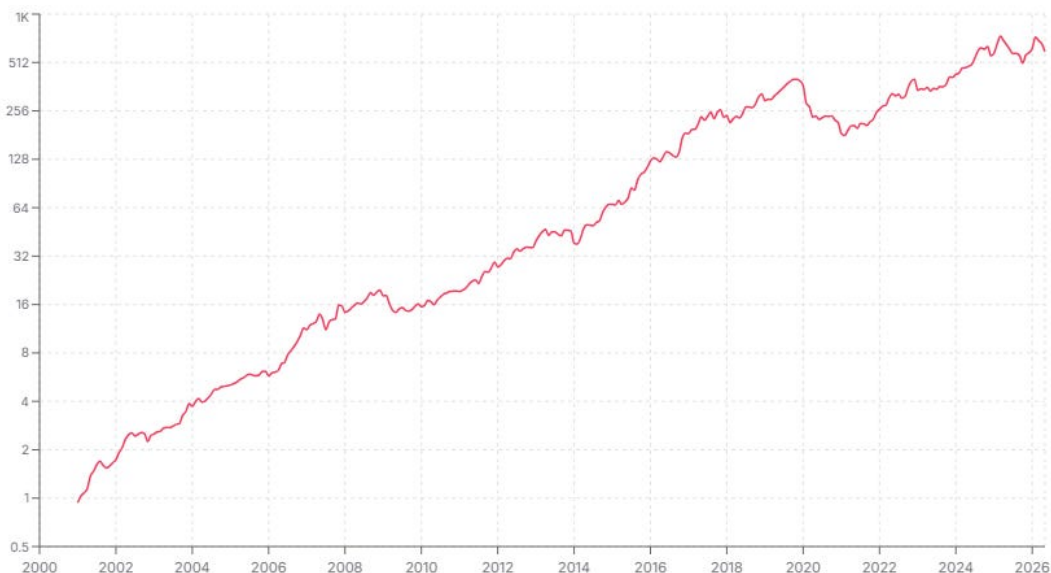
Using a **value-weighted** construction, the **highest-volatility decile (D10)** has returned an impressive **+83% YTD (through the end of May)**, while the **lowest-volatility decile (D1)** has declined **-5.3%**. Consequently, a traditional value-weighted low-volatility long-short portfolio has experienced a **significant drawdown** this year.



Equity curve of a value-weighted long-short low-volatility portfolio (Long D1 / Short D10), with both legs independently scaled to a 20% annualized volatility target.
Chart generated by [Concretum Factor Tracker](#).

This divergence is largely a consequence of the exceptional market concentration observed in 2026. Value-weighted portfolios naturally concentrate capital in the largest companies, leaving them heavily exposed to the extraordinary performance of a handful of high-volatility mega-cap technology and semiconductor stocks that have dominated this year's AI-driven rally. Meanwhile, many of the largest low-volatility mega-cap companies have lagged the broader market. **The resulting performance gap between the two deciles is therefore substantially larger than under an equally weighted construction.**

In fact, the picture is noticeably different using the **equally weighted** portfolios shown throughout this article. In this case, the **highest-volatility decile** is up **+17.2% YTD**, while the **lowest-volatility decile** has gained **+5.4%**. An equally weighted long-short implementation with volatility targeting approach is therefore **down only about 1% YTD**, despite one of the strongest rallies in the most volatile growth stocks in recent years.



Equity curve of an equally weighted long-short low-volatility portfolio (Long D1 / Short D10), with both legs independently scaled to a 20% annualized volatility target. Chart generated by [Concretum Factor Tracker](#).

This highlights an important lesson: portfolio construction matters. Choices such as **equal versus value weighting, volatility scaling, and position concentration** can materially affect short-term performance, even when the underlying long-term anomaly remains intact.

It also raises a broader question:

Should investors maintain unconditional exposure to a factor even when its performance trend has deteriorated?

In a recent article, we explored this idea by applying **cross-sectional and time-series momentum techniques across equity factors**. A momentum-based allocation would likely have reduced or exited exposure to the low-volatility factor relatively early during its 2026 deterioration, potentially limiting the impact of the drawdown.

For readers interested in this approach, see the following article where we examine how momentum can be used to allocate dynamically across equity-factor premia.



Timing Equity Factors with Momentum

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More detailed and up-to-date comparisons between equal-weighted and value-weighted factor portfolios are available through **Concretum Factor Tracker**.

Note. While the equally weighted implementation has proved more resilient during 2026, this should not be interpreted as evidence that it is universally superior. Equal weighting typically increases exposure to smaller and less liquid stocks, making the strategy less scalable for larger portfolios. By contrast, **value-weighted implementations are generally more practical at institutional scale**, although they may become more sensitive to periods of extreme mega-cap concentration.

Conclusion

The low-volatility anomaly is probably one of the most interesting contradictions in finance. According to the CAPM, investors should earn higher expected returns by taking more risk. However, decades of empirical evidence suggest exactly the opposite: the more risk you take, the lower your long-term returns tend to be.

Of course, this does not mean that investing in low-volatility stocks is a free lunch. Like any other equity factor, it can go through long periods of weak performance, sometimes for several years. As we have seen during 2026, portfolio construction, market concentration, and factor timing can have a much larger impact on short-term results than many investors expect, even if the long-term premium is still there.

If you would like to explore the anomaly in greater depth, we strongly suggest to read **Pim van Vliet's excellent book *High Returns from Low Risk***.

All the charts used in this article were generated using Factor Tracker, a FREE tool developed by the our team in Concretum.



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